Week 9 Development (Session 4)

1.Create Git account and configure repository

2.Upload the artifacts created to Git

1.Create Git account and configure repository

Git Tool

- Git authored by Linus Torvals in 2005.
- Git is a software for tracking changes in any set of files.
- Git is used for coordinating work among programmers collaboratively developing source code during software development .
- Git is a distributed version control system.

Features of Git

- High Speed.
- Data integrity.
- Support for distributed and non-linear workflows.
- Thousands of parallel branches can run on different systems.
- Every Git directory on every computer is a full-fledged repository with complete history & full version tracking abilities.
- This is independent of network access or a Central Server.
- Git is free and open Source Software distributed under the GPL-2.0-only License.

1.Create Git account and configure repository

Installation of Git on Windows

- Start downloading & Installing Git on system.
- Git bash integrate.
- Basic Git Commands.
- Create a Local Repository, that is Repository on a local machine.
- Connect to Remote Repository.

- Push file on Git house.

Step 1:

- Download Git for Windows(https://git-Scm.com/download/win)
- Click on Git-Downloading Package Select the below link
- https://git-scm.com › download › win.%C2%A0
- Download for Windows. Click here to download the latest (2.37.0) 32-bit version of Git for Windows.

Step 2:

- After downloading install, using normal steps Run, Next & install.
- Launch the Git Bash.
- Git Bash interface Occurs which is a black screen.

Step 3:

- In the Git Bash interface basic git commands can be worked on.

```
ADMIN@DELL MINGW64 ~/test (master)
$

ADMIN@DELL MINGW64 ~
$ git--version
bash: git--version: command not found

check for the version

ADMIN@DELL MINGW64 ~
$ git --version
git version 2.19.1.windows.1

ADMIN@DELL MINGW64 ~
$ git -version
```

```
Next explore the help command type git help config

$ git --help
usage: git [--version] [--help] [-C <path>] [-c <name>=<value>]
           [--exec-path[=<path>]] [--html-path] [--man-path] [--info-path]
           [-p | --paginate | -P | --no-pager] [--no-replace-objects] [--bare]
           [--git-dir=<path>] [--work-tree=<path>] [--namespace=<name>]
           <command> [<args>]

These are common Git commands used in various situations:

start a working area (see also: git help tutorial)
   clone      Clone a repository into a new directory
```

```
   init       Create an empty Git repository or reinitialize an existing one

work on the current change (see also: git help everyday)
   add        Add file contents to the index
   mv         Move or rename a file, a directory, or a symlink
   reset      Reset current HEAD to the specified state
   rm         Remove files from the working tree and from the index

examine the history and state (see also: git help revisions)
   bisect     Use binary search to find the commit that introduced a bug
   grep       Print lines matching a pattern
   log        Show commit logs
   show       Show various types of objects
   status     Show the working tree status

grow, mark and tweak your common history
   branch     List, create, or delete branches
   checkout   Switch branches or restore working tree files
   commit     Record changes to the repository
   diff       Show changes between commits, commit and working tree, etc
   merge      Join two or more development histories together
   rebase     Reapply commits on top of another base tip
   tag        Create, list, delete or verify a tag object signed with GPG

collaborate (see also: git help workflows)
   fetch      Download objects and refs from another repository
   pull       Fetch from and integrate with another repository or a local branch
   push       Update remote refs along with associated objects

'git help -a' and 'git help -g' list available subcommands and some
concept guides. See 'git help <command>' or 'git help <concept>'
to read about a specific subcommand or concept.
```

```
Create a new local directory(any name can be given for the directory)
 $ mkdir test

test is a new directory

ADMIN@DELL MINGW64 ~
$ mkdir test

ADMIN@DELL MINGW64 ~
$ cd test

ADMIN@DELL MINGW64 ~/test
$


ADMIN@DELL MINGW64 ~/test
$ git init
Initialized empty Git repository in C:/Users/ADMIN/test/.git/

ADMIN@DELL MINGW64 ~/test (master)
```

- Check for the new directory In Local: Disk C.
- We find users in Local Disk C.
- In users go to Admin.
- In admin test directory is found which is empty.

- The status is checked using below command

```
ADMIN@DELL MINGW64 ~/test (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)

        New folder/

nothing added to commit but untracked files present (use "git add" to track)
```

- **Now in this test directory create a text document with any name "demo" and type in say 'hello simplylearn' and then check for the git status.**

```
ADMIN@DELL MINGW64 ~/test (master)
$ git add demo.txt

ADMIN@DELL MINGW64 ~/test (master)

```

```
ADMIN@DELL MINGW64 ~/test (master)
$ git commit -m"commiting a txt file"

*** Please tell me who you are.

Run

  git config --global user.email "you@example.com"
  git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

```

- **A git hub account needs to be created.**
- **Link git to git hub account, the command for doing that is git config --global user.username.**
- **https//github.com sign up**
- **will ask for email and password, the password could be different than you mail id.**
- **Password should be alphanumeric and strong.**
- **Finally will ask for username. The same user name should be given in the bash screen.**

```
ADMIN@DELL MINGW64 ~/test (master)
$ git config --global user.usermame lms22se
.. ADMIN@DELL MINGW64 ~/test (master)
$ git remote add origin https://github.com/lms22se/test_demo.git

ADMIN@DELL MINGW64 ~/test (master)
$
```

- **Local and remote repository are linked by using git remote add origin.**
- **A new repository is added by clicking on +symbol and given name test_demo.**

- **click on create.**
- **Then the http command is copied on to bash screen.**
- **Push the local file on remote repository.**
- **Git push origin master command is used.**
- **Finally repository is created in your Git Hub with name Test.**

2.Upload the Artifacts created to Git

Artifact

- An artifact is a byproduct of software development that helps describe the architecture, design and function of software.
- Artifacts are like roadmaps that software developers can use to trace the entire software development process.
- Artifacts might be databases, datamodels, printed documents or scripts.

Types of Artifact

- **Code Artifacts**: Every program is made up of codes, and each coded process produces artifacts. Code artifacts are unavoidable, important by-products of the coding process.
- **Documentation**: Essentially, any piece of documentation relating to the software at hand is a relevant artifact.
- **Project Management Artifacts**: These artifacts come in the project management phase. They may come after routine tests of software, or after bug-checks.

Step 1:

- In the previous experiement already a link is make between Git and Git Hub.
- Also a repository also has been created in git hub.

Step 2

- Upload artifact action consists of two steps.
- Share data between jobs.
- Store build artifacts after a workflow is completed.

Step 3

- Git hub Actions loads the workflow, these workflow will not be able to pass files from one job to another.
- Artifact Action helps to take files & load into your workflow, these files are accessed by other jobs in workflow & examined after workflow is finished.

Step 4

- The easiest way to use Artifact Action in git hub is by storing these actions in the workflows .
- Then these actions can be used by other jobs after the workflows have finished.

Step 5

- Finally Upload Artifact action is used to upload files.
- Github.com/Action/upload-Artifact.
- To view the same we can download it.